

AWS SOC Monitoring - CreateUser Detection Workflow

Step 1: Simulate Suspicious Activity

Create a new IAM user inside AWS IAM service. This generates a CreateUser API event in AWS CloudTrail logs.

The screenshot shows the 'Specify user details' step of the AWS IAM 'Create user' process. The left sidebar indicates the current step is 'Specify user details' (Step 1), with other steps being 'Set permissions', 'Review and create', and 'Retrieve password'. The main content area is titled 'Specify user details' and contains the following sections:

- User details:**
 - User name:** A text input field containing 'Security-engineer'. Below it, a note states: 'The user name can have up to 64 characters. Valid characters: A-Z, a-z, 0-9, and +, -, @, _ (hyphen)'.
 - Provide user access to the AWS Management Console - optional:** A checkbox that is checked. A note below states: 'In addition to console access, users with `SignInLocalDeveloperAccess` permissions can use the same console credentials for programmatic access without the need for access keys.'
- Console password:**
 - Autogenerated password:** A radio button that is unselected. A note below states: 'You can view the password after you create the user.'
 - Custom password:** A radio button that is selected. A text input field contains a masked password '*****'. Below it, a note states: 'Enter a custom password for the user.'
 - Must be at least 8 characters long** and **Must include at least three of the following mix of character types: uppercase letters (A-Z), lowercase letters (a-z), numbers (0-9), and symbols ! @ # \$ % ^ & * () _ + - (hyphen) = [] { }**
 - Show password:** A checkbox that is unselected.
- Users must create a new password at next sign-in - Recommended:** A checkbox that is checked. A note below states: 'Users automatically get the `IAMUserChangePassword` policy to allow them to change their own password.'

At the bottom right, there are 'Cancel' and 'Next' buttons.

The screenshot shows the 'Set permissions' step of the AWS IAM 'Create user' process. The left sidebar indicates the current step is 'Set permissions' (Step 2), with other steps being 'Specify user details', 'Review and create', and 'Retrieve password'. The main content area is titled 'Set permissions' and contains the following sections:

- Permissions options:** Three radio buttons are present:
 - Add user to group:** Unselected. A note below states: 'Add user to an existing group, or create a new group. We recommend using groups to manage user permissions by job function.'
 - Copy permissions:** Unselected. A note below states: 'Copy all group memberships, attached managed policies, and inline policies from an existing user.'
 - Attach policies directly:** Selected. A note below states: 'Attach a managed policy directly to a user. As a best practice, we recommend attaching policies to a group instead. Then, add the user to the appropriate group.'
- Permissions policies (1/1493):** A section with a search bar containing 'ec2' and a 'Filter by Type' dropdown set to 'All types'. It shows 50 matches. Below is a table of policies:

Policy name	Type	Attached entities
☐ AmazonEC2ContainerRegistryFullAccess	AWS managed	0
☐ AmazonEC2ContainerRegistryPowerUser	AWS managed	0
☐ AmazonEC2ContainerRegistryPullOnly	AWS managed	0
☐ AmazonEC2ContainerRegistryReadOnly	AWS managed	0
☐ AmazonEC2ContainerServiceAutoscaleRole	AWS managed	0
☐ AmazonEC2ContainerServiceEventsRole	AWS managed	0
☐ AmazonEC2ContainerServiceFortiC2Role	AWS managed	0
☐ AmazonEC2ContainerServiceRole	AWS managed	0
☒ AmazonEC2FullAccess	AWS managed	1

Step 1

Specify user details

Step 2

Set permissions

Step 3

Review and create

Step 4

Retrieve password

Review and create

Review your choices. After you create the user, you can view and download the autogenerated password, if enabled.

User details

User name

Security-engineer

Console password type

Custom password

Require password reset

Yes

Permissions summary

< 1 >

Name ↗

▲

Type

▼

Used as

▼

[AmazonEC2FullAccess](#)

AWS managed

Permissions policy

[IAMUserChangePassword](#)

AWS managed

Permissions policy

Tags - optional

Tags are key-value pairs you can add to AWS resources to help identify, organize, or search for resources. Choose any tags you want to associate with this user.

No tags associated with the resource.

Add new tag

You can add up to 50 more tags.

Cancel

Previous

Create user

✔ User created successfully

You can view and download the user's password and email instructions for signing in to the AWS Management Console.

View user

Step 1

Specify user details

Step 2

Set permissions

Step 3

Review and create

Step 4

Retrieve password

Retrieve password

You can view and download the user's password below or email users instructions for signing in to the AWS Management Console. This is the only time you can view and download this password.

Console sign-in details

Email sign-in instructions ↗

Console sign-in URL

<https://922806890303.signin.aws.amazon.com/console>

User name

[Security-engineer](#)

Console password

[XXXXXXXXXX](#) [Show](#)

Cancel

Download .csv file

Return to users list

Step 2: Verify CloudTrail Event

Verify that the CreateUser event is successfully recorded inside AWS CloudTrail Event History.

CloudTrail

Dashboard

Event coverage

Event history

Insights

Trails

Lake

Dashboards

Query

Event data stores

Integrations

Settings

Pricing

Documentation

Event history (0)

Info

Download events

Query in Lake

Create Athena table

Event history shows you the last 90 days of management events.

Lookup attributes

Event name

AdminCreateUser

Filter by date and time

Clear filter

1

Event name

Event time

User name

Event source

Resource type

Resource name

No matches

We can't find a match.

0 / 5 events selected

Search

[Alt+S]

United States (N. Virginia)

rahana (922806890303)

rahana

CloudTrail

Dashboard

Event coverage

Event history

Insights

Trails

Lake

Dashboards

Query

Event data stores

Integrations

Settings

Pricing

Event history (3)

Info

Download events

Query in Lake

Create Athena table

Event history shows you the last 90 days of management events.

Lookup attributes

Event name

CreateUser

Filter by date and time

Clear filter

1

☐

Event name

☐

CreateUser

☐

CreateUser

☐

CreateUser

May 15, 2026, 11:07:22 (UTC+0...)

May 10, 2026, 17:06:26 (UTC+0...)

May 10, 2026, 15:01:31 (UTC+0...)

root

root

root

iam.amazonaws.com

iam.amazonaws.com

iam.amazonaws.com

AWS::IAM::User, AWS::I...

AWS::IAM::User, AWS::I...

AWS::IAM::User, AWS::I...

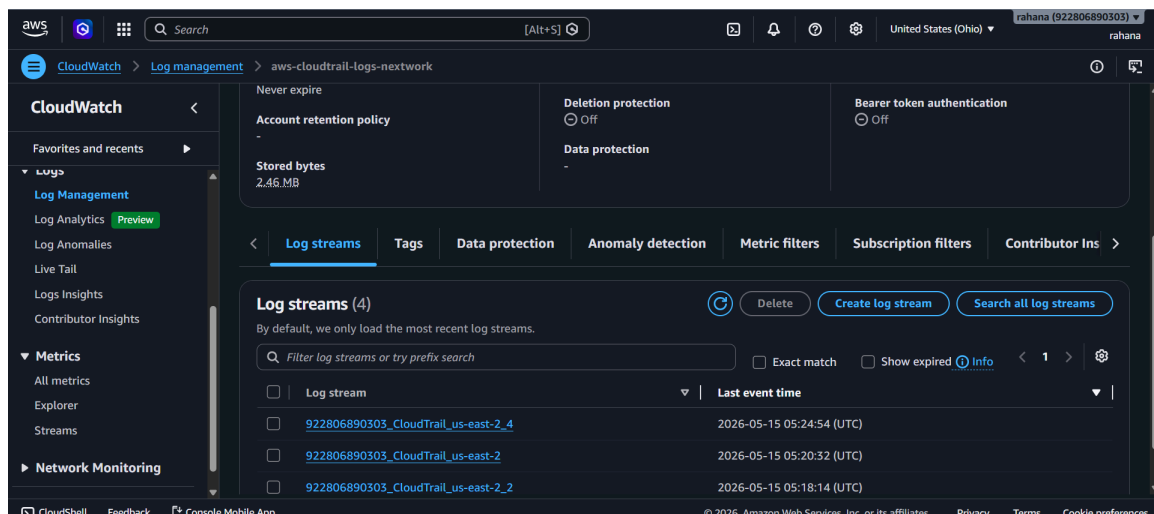
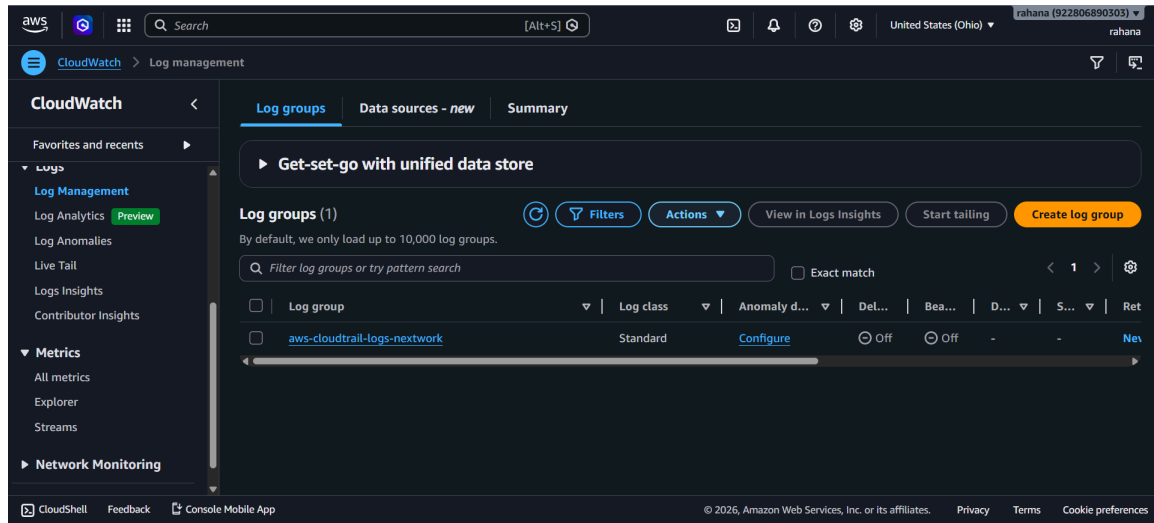
AIDASNW4TV47RWIMGIDCF, Security-engineer, am...

security-admin, arn:aws:iam:922806890303:user/s...

AIDA5NW4TV47VGIOHENUV, security-admin, amta...

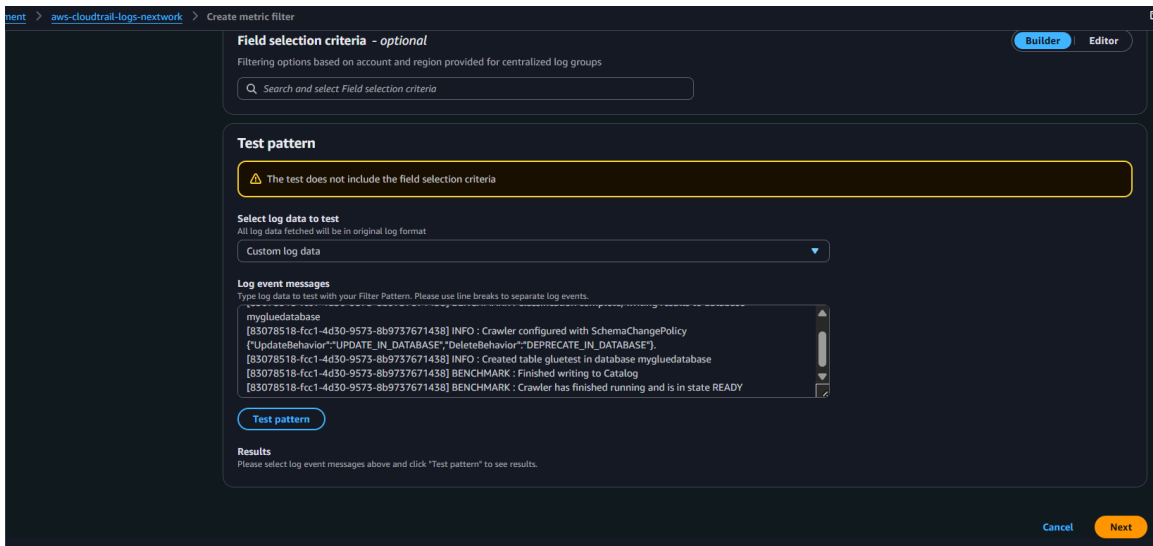
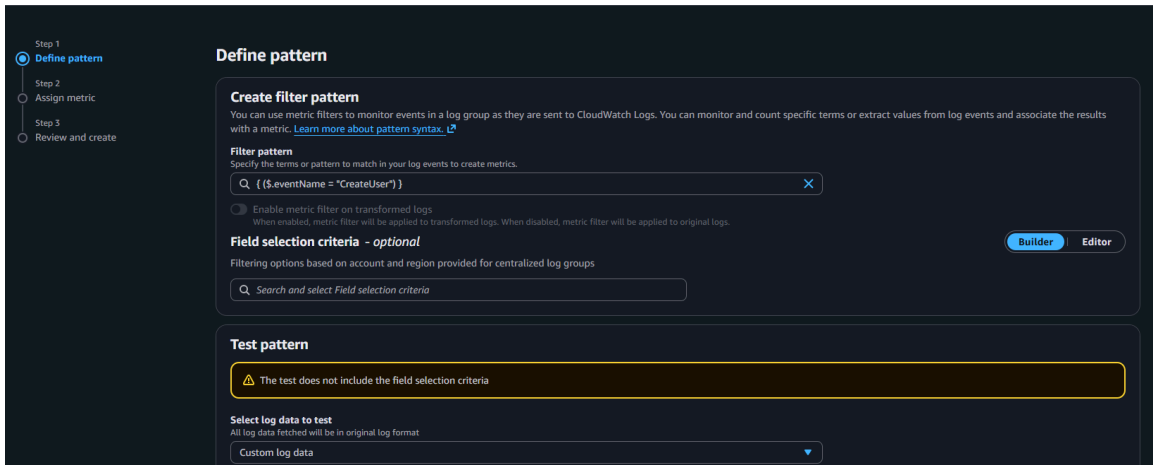
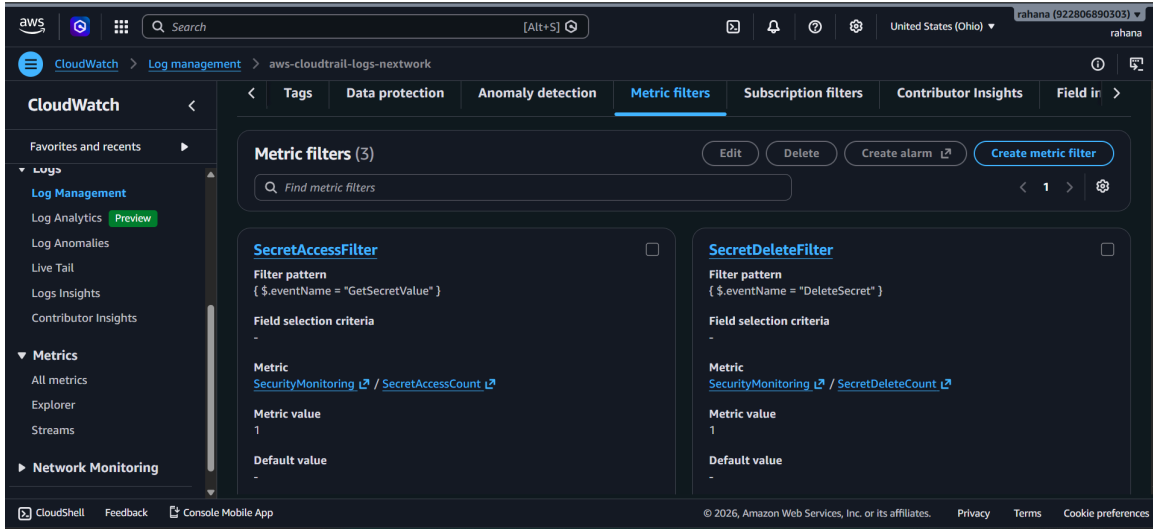
Step 3: Open CloudWatch Log Groups

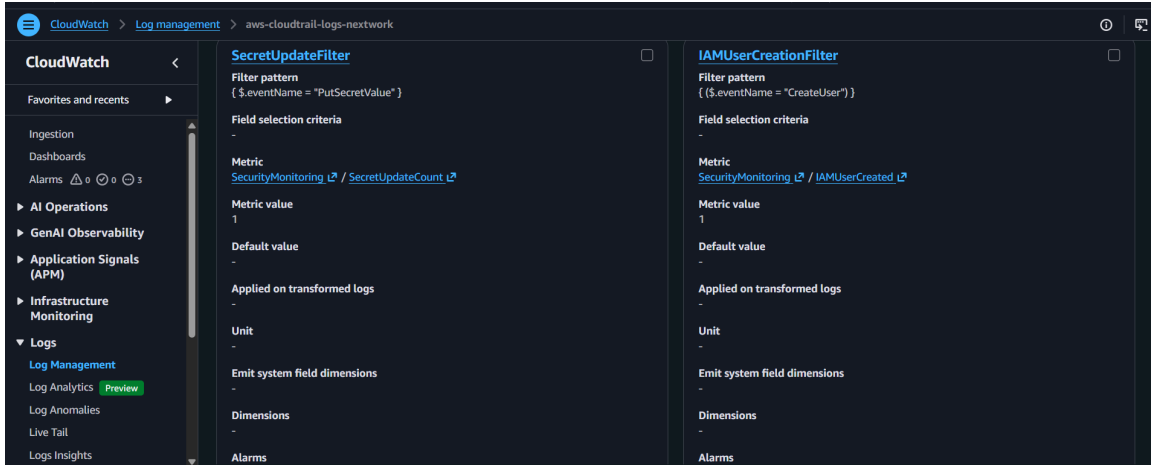
Access CloudWatch Log Groups connected to CloudTrail and verify that CreateUser logs are available in log streams.



Step 4: Create Metric Filter

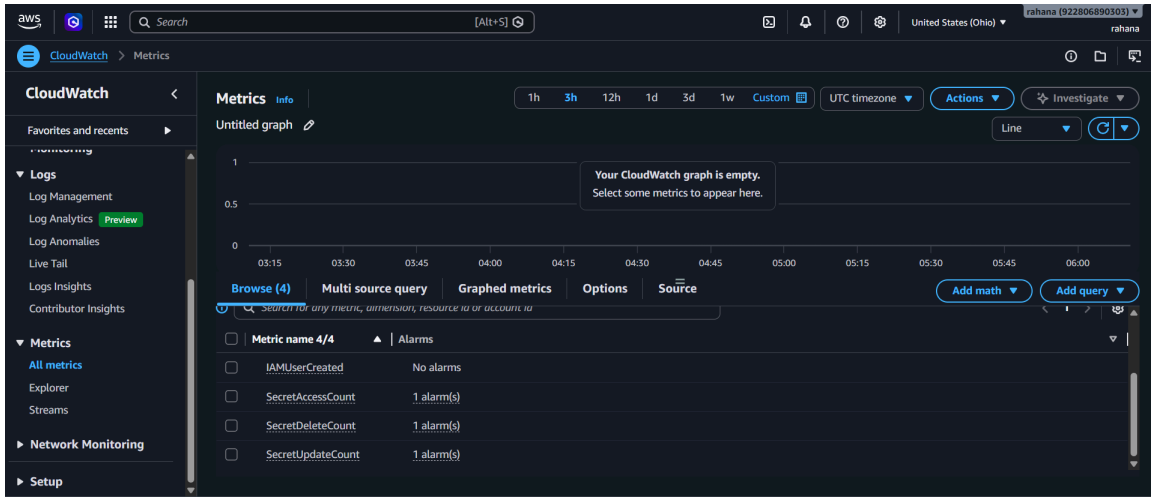
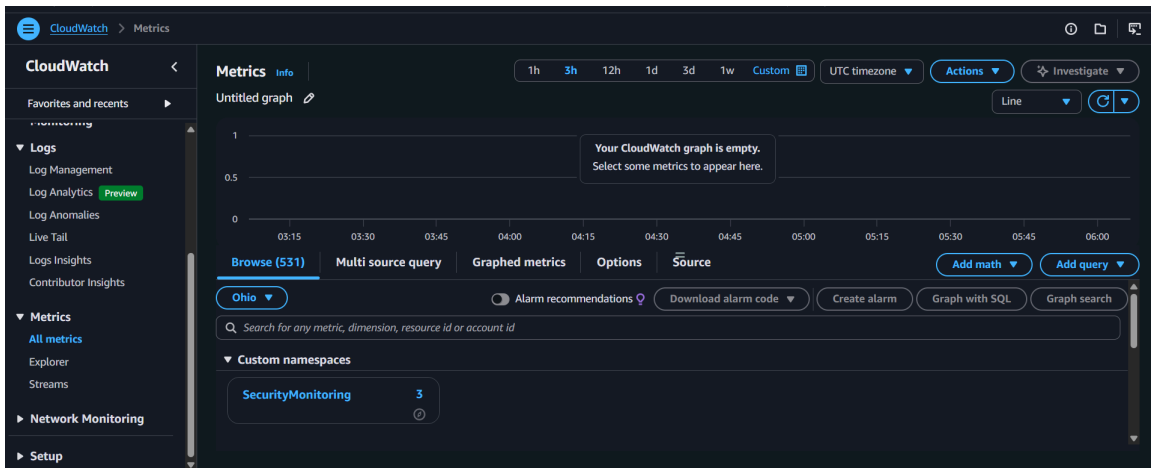
Create a CloudWatch Metric Filter using the pattern: `{ ($.eventName = "CreateUser") }` to detect IAM user creation events.





Step 5: Verify Metric Creation

Verify that the IAMUserCreated metric is successfully created inside the SecurityMonitoring namespace.



Step 6: Create CloudWatch Alarm

Create a CloudWatch Alarm with Statistic=Sum, Threshold ≥ 1 , and Period=5 minutes to detect CreateUser events.

The screenshot shows the AWS CloudWatch console interface. At the top, there's a navigation bar with tabs: "IAMUserCreated", "Browse (4)", "Multi source query", "Graphed metrics (1)", "Options", and "Source". Below the navigation bar, there are buttons: "Add math", "Add query", "Download alarm code", "Create alarm", "Graph with SQL", and "Graph search". A search bar contains the text "Search for any metric, dimension, resource id or account id". Below the search bar, there's a section titled "Metric name 4/4" with a dropdown menu showing "Ohio". Below this, there's a table with one row: "IAMUserCreated" with the value "No alarms".

The screenshot shows the "Specify metric and conditions" step in the AWS CloudWatch console. On the left, there's a sidebar with steps: "Step 1: Specify metric and conditions", "Step 2: Configure actions", "Step 3: Add alarm details", and "Step 4: Preview and create". The main content area is titled "Specify metric and conditions". It has two main sections: "Metric" and "Conditions".

Metric Section:

- Graph:** A placeholder for a graph showing "No data available. Try adjusting the dashboard time range." The x-axis shows time from 03:30 to 06:00. The y-axis shows the metric name "IAMUserCreated".
- Form fields:**
 - Namespace:** SecurityMonitoring
 - Metric name:** IAMUserCreated
 - Statistic:** Sum
 - Period:** 5 minutes

Conditions Section:

- Threshold type:** Static (selected) or Anomaly detection.
- Whenever IAMUserCreated is...** Define the alarm condition.
 - Greater:** > threshold
 - Greater/Equal:** $\geq$ threshold (selected)
 - Lower/Equal:** $\leq$ threshold
 - Lower:** < threshold
- than...** Define the threshold value.
 - Value:** 1
 - Must be a number.**
- Additional configuration:** (expandable section)

At the bottom right, there are "Cancel" and "Next" buttons.

Step 1

Specify metric and conditions

Step 2

Configure actions

Step 3

Add alarm details

Step 4

Preview and create

Add alarm details

Name and description

Alarm name

IAM-User-Creation-Alert

Alarm description - optional

View formatting guidelines

Edit

Preview

This is an H1

""double asterisks will produce strong character""

This is [an example](https://example.com/) inline link.

Up to 1024 characters (0/1024).

ⓘ

Markdown formatting is only applied when viewing your alarm in the console. The description will remain in plain text in the alarm notifications.

Tags - optional

Info

No tags associated with the resource.

Add new tag

You can add up to 50 tags.

Cancel

Previous

Next

Step 1

Specify metric and conditions

Step 2

Configure actions

Step 3

Add alarm details

Step 4

Preview and create

Preview and create

Step 1: Specify metric and conditions

Metric

Graph

This alarm will trigger when the blue line goes above the red line for 1 datapoints within 5 minutes.

No data available.

Try adjusting the dashboard time range.

1

03:30

04:00

04:30

05:00

05:30

06:00

IAMUserCreated

Namespace

SecurityMonitoring

Metric name

IAMUserCreated

Statistic

Sum

Period

5 minutes

Conditions

Threshold type

Static

Whenever IAMUserCreated is

Greater/Equal ($\geq$)

than...

1

Additional configuration

Step 2: Configure actions

Actions

Notification

When In alarm, send a notification to "SecurityAlertsbynetwork"

Step 3: Add alarm details

Alarm details

Name

IAM-User-Creation-Alert

Description

-

Tags (0)

Cancel

Previous

Create alarm

Successfully created alarm IAM-User-Creation-Alert.

View alarm

Alarms

Mute rules - new

Alarms (4)

Filter alarms

Quick filters

Clear selection

Create composite alarm

Actions

Create alarm

☐	Name	State	Actions	Actions muted - new	Last state update (UTC)	Conditions
☐	IAM-User-Creation-Alert	Insufficient data	Actions enabled	-	2026-05-15 06:16:10	IAMUserCreated $\geq$ 1 for 1 datapoints within 5 minutes

Step 9: Verify SNS Alert & Splunk Detection

Confirm SNS email alert delivery and validate CreateUser event ingestion into Splunk using SPL query: `index=* sourcetype=aws:cloudtrail eventName=CreateUser`

ALARM: "IAM-User-Creation-Alert" in US East (Ohio) [Summarize this email](#)



AWS Notifications <no-reply@sns.amazonaws.com>

To: @ Nadil AN

Reply Reply all Forward ...
Fri 5/15/2026 11:47 AM

You are receiving this email because your Amazon CloudWatch Alarm "IAM-User-Creation-Alert" in the US East (Ohio) region has entered the ALARM state, because "Threshold Crossed: 1 out of the last 1 datapoints [1.0 (15/05/26 06:07:00)] was greater than or equal to the threshold (1.0) (minimum 1 datapoint for OK -> ALARM transition)."

View this alarm in the AWS Management Console:

<https://us-east-2.console.aws.amazon.com/cloudwatch/deeplink.js?region=us-east-2#alarmsV2:alarm/IAM-User-Creation-Alert>

Alarm Details:

- Name: IAM-User-Creation-Alert
- Description:
- State Change: INSUFFICIENT_DATA -> ALARM
- Reason for State Change: Threshold Crossed: 1 out of the last 1 datapoints [1.0 (15/05/26 06:07:00)] was greater than or equal to the threshold (1.0) (minimum 1 datapoint for OK -> ALARM transition).
- Timestamp: Friday 15 May, 2026 06:17:53 UTC
- AWS Account: 922806890303
- Alarm Arn: arn:aws:cloudwatch:us-east-2:922806890303:alarm:IAM-User-Creation-Alert

Threshold:

- The alarm is in the ALARM state when the metric is GreaterThanOrEqualToThreshold 1.0 for at least 1 of the last 1 period(s) of 300 seconds.

Monitored Metric:

- MetricNamespace: SecurityMonitoring
- MetricName: IAMUserCreated
- Dimensions:
- Period: 300 seconds
- Statistic: Sum
- Unit: not specified
- TreatMissingData: missing

State Change Actions:

- OK:
- ALARM: [arn:aws:sns:us-east-2:922806890303:SecurityAlertsbynexwork]
- INSUFFICIENT_DATA:

splunk-enterprise Apps Administrator Messages Settings Activity Help Find

Search Analytics Datasets Reports Alerts Dashboards Modules

New Search

Index= default eventName= CreateUser

Time range: All time

0 events (before 5/15/26 12:01:07:000 PM) No Event Sampling

Events (0) Patterns Statistics Visualization

No results found.

New Search

Index= sourcetype=aws:cloudtrail

Time range: All time

1,010 events (before 5/15/26 12:02:23:000 PM) No Event Sampling

Events (1,010) Patterns Statistics Visualization

Timeline format Zoom Out Zoom to Selection Deselect 1 hour per column

Time	Event
5/15/26 11:58:28.000 AM	{ [] additionalEventData: [[] awsRegion: us-east-2 eventCategory: Management eventID: 9bc54605-3446-376d-9363-e8fab82a8cc eventName: AssumeRole eventSource: sts.amazonaws.com eventTime: 2026-05-15T06:28:28Z eventType: AwsApiCall eventVersion: 1.11 managementEvent: true readOnly: true recipientAccountId: 922806890303 requestId: 87016632-86a2-429f-a85f-a84c71f53f7a requestParameters: [[]] }

Conclusion

Successfully implemented AWS SOC monitoring for IAM user creation activities using CloudTrail, CloudWatch Metric Filters, CloudWatch Alarms, SNS notifications, S3 log storage, SQS integration, and Splunk SIEM. This workflow demonstrates real-time detection engineering, alerting, and centralized security monitoring for suspicious AWS IAM activities.